

Prepify — Final Defense Presentation

Project: Prepify — A Community Driven Platform for Handwritten Academic Notes **Student:** Shorno Kamal Roy (222-15-6141) **Supervisor:** Md. Sazzadur Ahamed, Assistant Professor **Co-Supervisor:** Md. Ashraful Islam Talukder, Lecturer **Department:** CSE, Daffodil International University **Date:** May 2026

Total slides: 17 (12–15 minutes presentation + 3–5 minutes Q&A) **Focus area:** Novelty of the Work (Slide 11)

Common Visual Style Preface for All AI-Generated Slide Visuals

Append this to the front of any new visual prompt for visual consistency across the deck. It differs slightly from the report-diagram preface — presentation visuals need bolder, simpler, more presentation-friendly designs:

Modern presentation visual, 16:9 aspect ratio, large legible elements designed for projection, white or very light gray (#F8FAFC) background, flat vector illustration style, no gradients, no 3D, no drop shadows, sharp clean edges, generous whitespace, sans-serif typography (Inter, Helvetica, or SF Pro), large readable type sizing (minimum 24 pt for body text and 36 pt for titles when scaled to 1920×1080), bold accent colors limited to this palette: deep slate #0F172A for primary text, blue #2563EB for client / user side, indigo #4F46E5 for server / application logic, emerald #059669 for data / persistence, amber #F59E0B for third-party services, rose #E11D48 for AI / novel features, light gray #F1F5F9 for grouping containers, red #DC2626 for problems / negatives, green #16A34A for solutions / positives. No clip art, no decorative people figures, no stock-photo gradients. Every text label must be horizontal and immediately readable.

SLIDES

Slide 1 — Title Slide

On slide:

- **Prepify**
- A Community Driven Platform for Handwritten Academic Notes
- Final Year Design Project — Phase II Defense
- **Shorno Kamal Roy** — 222-15-6141
- **Supervisor:** Md. Sazzadur Ahamed, Assistant Professor
- **Co-Supervisor:** Md. Ashraful Islam Talukder, Lecturer
- Department of CSE, Daffodil International University
- May 2026

Visual prompt (Slide 1):

16:9 title slide for an academic project defense. Clean white background with a subtle light-gray decorative element (geometric, not floral) in one corner. Centered: large bold title "Prepify" in indigo

#4F46E5 (96 pt equivalent), subtitle "A Community Driven Platform for Handwritten Academic Notes" in slate #0F172A (40 pt), tag line "Final Year Design Project — Phase II Defense" in muted gray (28 pt). Bottom-left corner: student name and ID in two lines. Bottom-right corner: small Daffodil International University text logo. Top-right: small "May 2026" date stamp. No icons, no background images, professional academic aesthetic.

Slide 2 — Introduction

On slide:

- Handwritten notes are the primary study material in Bangladeshi universities
- Currently shared informally via WhatsApp, Telegram, Facebook groups
- Three persistent pain points:
 1. Notes get **buried** under unrelated chats
 2. **No search**, no tagging, no organization
 3. **No quality signal** — good and bad notes look identical
- **Prepify** turns this fragmented behaviour into a structured, free, AI-assisted platform

Speaker notes: "Every semester, students scramble to collect handwritten notes from seniors. They post requests in WhatsApp groups and scroll through hundreds of messages. The notes already exist — what doesn't exist is a place to organize, rank, and reuse them. Prepify is built around this insight."

Visual prompt (Slide 2 — Problem Comparison):

Side-by-side 16:9 comparison illustration titled "From Chaos to Structure". Left half (50% width, light red #FEF2F2 background, red #DC2626 border on top with "BEFORE" label): a stylized illustration of a messy phone chat interface showing a chaotic stream of overlapping chat bubbles labeled "anyone has CSE 313 notes?", "midterm tomorrow plz help", "who has math notes from sir Karim?", "lost the link", "send again pls", with three of the bubbles partially covered by newer ones to imply burial. Add three small red icons labeled "Buried", "Unsearchable", "No Quality Signal" running down the left edge. Right half (50% width, light green #F0FDF4 background, green #16A34A border on top with "AFTER" label): a stylized illustration of a clean Prepify-like card grid showing 6 organized note cards in a 2x3 layout, each card with a course tag (CSE 313, MAT 102, EEE 211, etc.), a like count badge, and a small avatar. Three small green icons labeled "Organized", "Searchable", "Community-Ranked" running down the right edge. Center divider: a vertical arrow pointing right with the word "Prepify" inside an indigo #4F46E5 rounded badge. No gradients, flat vector style.

Slide 3 — Objectives

On slide:

1. **Build the core platform.** A free, structured space for uploading and discovering handwritten notes — organized by Faculty, Department, and Course, with role-based moderation.
2. **Build the engagement layer.** Likes, comments, follows, leaderboard, streaks, badges, and curated collections — the social loop that surfaces quality.

3. Integrate multimodal AI. Gemini 2.5 Flash reads handwritten images directly and returns structured explanations — no OCR step.

Speaker notes: "Three objectives, one for each layer of the system — the platform, the engagement loop on top of it, and the AI pipeline that sets it apart."

Visual prompt (Slide 3 — Three-Objective Layout):

16:9 slide titled "Objectives" in slate top-left. Three objective panels stacked vertically with generous spacing (each panel ~25% of slide height, 16 px gap). Each panel is a rounded rectangle (16 px corners) with white fill, 2 px slate border, 28 px internal padding, and a colored 8-px-thick left border. Panel layout left to right: (a) 88-px-wide left column with a large filled colored circle containing a bold white "1", "2", or "3" (36 pt), (b) wide content area with a bold short title (24 pt) and a single one-line description below it (16 pt muted slate), (c) small monoline icon at the far right.

Panels in order:

- 1. Core Platform** — indigo #4F46E5. Description: "Free, structured space for handwritten notes with role-based moderation." Icon: stacked-folders.
- 2. Engagement Layer** — amber #F59E0B. Description: "Likes, comments, leaderboard, streaks, badges, and curated collections." Icon: connected-people.
- 3. Multimodal AI** — rose #E11D48. Description: "Gemini 2.5 Flash reads handwritten images directly — no OCR." Icon: sparkles-on-document.

Bottom of slide: thin one-line banner in muted gray italic: "Three objectives, three layers of the system."
Flat vector, no gradients.

Slide 4 — Background Study (Part 1: Academic Literature)

On slide:

Author	Title	Objectives	Methodology	Key Findings
Chatti et al. (2010)	The Future of E-Learning	Define the Personal Learning Environment (PLE) concept	Conceptual framework + literature synthesis	Learners benefit from personalised, networked spaces over rigid institutional systems
Attwell (2007)	Personal Learning Environments — the future of eLearning?	Position PLE as the next stage of eLearning	Position paper with case observations	Learner control + trusted communities raise engagement and resource quality
Getenet et al. (2024)	Students' Digital Attitude & Online Learning Engagement	Measure how digital habits affect engagement	Quantitative survey of university students	Engagement is highest when a platform fits students' existing habits

Author	Title	Objectives	Methodology	Key Findings
Vaswani et al. (2017)	Attention Is All You Need	Replace recurrence with pure self-attention	Novel architecture + WMT benchmark experiments	Transformer architecture — the basis of every modern multimodal LLM
Google DeepMind (2023)	Gemini: A Family of Multimodal Models	Build natively multimodal foundation models	Large-scale joint training on text / image / audio / video	One model reads images (incl. handwriting & Bangla) and produces text in a single pass
Smith (2007)	An Overview of the Tesseract OCR Engine	Document Tesseract's architecture and capabilities	Engine description + recognition benchmarks	Strong on printed text; struggles on irregular handwriting

Speaker notes: "The literature splits into two threads. The top three rows are about *how* students learn best — personalised, community-driven, habit-aligned. The bottom three are about *what* technology can finally do — multimodal LLMs read handwriting directly, which is something Tesseract-style OCR was never reliable at. Prepify sits at the intersection of those two threads."

Visual prompt (Slide 4 — Literature Table):

16:9 slide titled "Background Study — Academic Literature" in slate top-left. A clean 6-row 5-column table fills the lower 82% of the slide. Header row in indigo #4F46E5 background with white bold 14 pt text. Column widths: Author 14%, Title 22%, Objectives 20%, Methodology 20%, Key Findings 24%. Body rows alternate white and very light slate #F8FAFC for readability, body text 11 pt slate, line height 1.4, vertical-aligned to top, 8 px cell padding. Author column shows author name in bold with the year in muted gray on the line below. Add a tiny 12-px circular accent dot to the left of each author name colored by category: blue #3B82F6 for platform/community/learning rows (Chatti, Attwell, Getenet), rose #E11D48 for AI/model rows (Vaswani, Google DeepMind), amber #F59E0B for legacy OCR (Smith). Allow long cells to wrap onto 2 lines. Bottom of the slide: one-line italic banner in light gray — "Two threads converge: community-driven learning + multimodal AI for handwriting." Flat vector, no gradients, no drop shadows.

Slide 5 — Background Study (Part 2: Existing Platforms)

On slide:

Platform	Approach	Limitation
Doccity	Global, freemium, AI in paid app	Typed content, paywall, no Bangladesh taxonomy
Studocu	Freemium typed documents	No handwriting focus
Thinkswap	Credit-based exchange	Must upload to download

Platform	Approach	Limitation
Course Hero / Chegg	Paid subscription	Western pricing
Ocenot (Bangladesh)	Paid marketplace	Site offline, no AI features

Visual prompt (Slide 5 — Competitor Capability Matrix):

16:9 slide titled "Existing Platforms — Capability Matrix". A capability matrix in the center with 6 rows (one per competitor + Prepify at the bottom) and 5 columns. Column headers (top row, indigo #4F46E5 background, white text, vertically rotated 30 degrees if needed): "Free Access", "Handwriting Focus", "Community Ranking", "Multimodal AI", "Bangladesh Taxonomy". Row headers (left column, slate background, white text): "Docsity", "Studocu", "Thinkswap", "Course Hero / Chegg", "Ocenot", "Prepify". Cell contents: large red ✖ (#DC2626) for missing capability, large green ✔ (#16A34A) for present capability, amber ⚠ (#F59E0B) for partial capability. Specifically: Docsity row → ✖, ✖, ⚠, ⚠, ✖; Studocu row → ✖, ✖, ✖, ✖, ✖; Thinkswap row → ✖, ✖, ✖, ✖, ✖; Course Hero row → ✖, ✖, ✖, ⚠, ✖; Ocenot row → ✖, ⚠, ✖, ✖, ✔; Prepify row → ✔, ✔, ✔, ✔, ✔ (the entire Prepify row is highlighted with a soft indigo glow background to draw the eye). At the very bottom, a one-line takeaway banner: "No existing platform fills all five columns." No gradients, flat presentation style.

Slide 6 — Gap Analysis

On slide:

Five gaps in the current landscape:

#	Gap	How Prepify Addresses It
1	No fully free access	No paywall, no credits, no premium tier
2	No handwriting-first platform	Upload flow + AI built around photographed pages
3	No Bangladeshi academic taxonomy	Faculty / Dept / Batch / Course tables modeled locally
4	AI features siloed in paid apps	Multimodal AI integrated into the note workflow
5	No engagement / gamification layer	Leaderboard, streaks, badges, public collections

Visual prompt (Slide 6 — Gap-to-Solution Cards):

16:9 slide titled "Gap Analysis". Five horizontal "gap → solution" cards stacked vertically with 12 px gap between each. Each card spans full slide width and is divided into three regions left to right: (a) a 60-px-wide vertical strip on the far left containing a large white number (1-5) on a slate #0F172A background, (b) a left content half titled "GAP" with red #DC2626 left border, light red #FEF2F2 fill, displaying the gap statement in dark red bold text, (c) a center 48-px-wide arrow-bridge zone showing a clean rightward arrow in indigo #4F46E5, (d) a right content half titled "PREPIFY'S APPROACH" with green #16A34A left border, light green #F0FDF4 fill, displaying the solution in dark green text. The five card pairs are:

- 1. GAP: No fully free access → SOLUTION: No paywall, no credits, no premium tier
- 2. GAP: No handwriting-first platform → SOLUTION: Upload flow + AI built around photographed pages
- 3. GAP: No Bangladeshi academic taxonomy → SOLUTION: Faculty / Department / Batch / Course tables modeled locally
- 4. GAP: AI siloed in paid apps → SOLUTION: Multimodal AI integrated into the note workflow
- 5. GAP: No engagement / gamification → SOLUTION: Leaderboard, streaks, badges, public collections

Flat vector, no gradients. Title at top in 36-pt slate.

Slide 7 — Methodology (Development Approach)

On slide:

- **Agile** — 1–2 week sprints over **32 weeks** (2 semesters)
- **Phase 1 (W1–16):** Auth, notes, voting, comments, follows, moderation, dashboards
- **Phase 2 (W17–32):** Bookmarks, streaks, badges, **multimodal AI pipeline**
- Single solo developer, weekly supervisor review
- Source control: Git + GitHub from day one

Visual: Reuse Figure 3.5 (Agile Cycle + Phase Timeline) from `Diagram_Generation_Prompts.md`. No new visual needed.

Slide 8 — Methodology (System Architecture)

On slide:

- **Three-tier architecture** — browser, Next.js application, data layer
- **Frontend:** Next.js 16.1, React 19.2, Tailwind 4, shadcn/ui
- **Backend:** Next.js Server Actions (no separate API server)
- **Database:** PostgreSQL on Neon (serverless), Drizzle ORM
- **External:** Cloudinary (images), Google OAuth (auth), Brevo (email), **Gemini 2.5 Flash** (AI)
- **Hosted on:** Hostinger VPS, Ubuntu 22.04, PM2 + Nginx + Let's Encrypt

Visual: Reuse Figure 3.1 (System Architecture) from `Diagram_Generation_Prompts.md`. No new visual needed.

Slide 9 — Results & Analysis (Features Delivered)

On slide:

22 production features across 7 modules:

Module	Features
Auth	Google OAuth, multi-step onboarding, role gating

Module	Features
Notes	Multi-image upload, course tagging, edit, delete
Engagement	Likes, comments, follows, notifications (in-app + email)
Gamification	Leaderboard, streaks, badges, points transaction log
Discovery	Bookmarks, public/private collections, dept browser
Moderation	Pending queue, approve/reject with reason, applications
AI	Multimodal explanations, regen cap, structured output

Visual prompt (Slide 9 — Module Mosaic):

16:9 slide titled "Features Delivered" with subtitle "22 features across 7 modules" in muted gray. Below the title, a 4×2 grid of "module tiles" (with the last cell of the 8-slot grid replaced by a summary count tile). Each tile is a rounded rectangle (16 px corners) with white fill and a 4-px colored top border. Inside each tile from top to bottom: a 36-px monoline icon, the module name in bold slate (24 pt), a horizontal divider line, then the comma-separated feature list in 16-pt muted slate. Color the top border of each tile by category:

- **Auth** (blue #2563EB): "Google OAuth, multi-step onboarding, role gating"
- **Notes** (indigo #4F46E5): "Multi-image upload, course tagging, edit, delete"
- **Engagement** (amber #F59E0B): "Likes, comments, follows, notifications (in-app + email)"
- **Gamification** (rose #E11D48): "Leaderboard, streaks, badges, points transaction log"
- **Discovery** (emerald #059669): "Bookmarks, public/private collections, dept browser"
- **Moderation** (slate #0F172A): "Pending queue, approve/reject with reason, applications"
- **AI** (rose #E11D48, larger and emphasized with a thicker border and slight scale-up to 105%): "Multimodal explanations, regen cap, structured output"

The 8th tile (bottom-right) is a "Total" summary tile in solid indigo #4F46E5 with white text showing "22 features" as a large number with "across 7 modules" as a subtitle below. Flat vector, no gradients.

Slide 10 — Results & Analysis (Performance & Testing)

On slide:

Manual testing — 139 test cases across 14 areas, all passed

Metric	Result
Lighthouse — Landing performance	87
Lighthouse — Accessibility	92
Initial HTML delivery	~800 ms
Time to interactive	~1.5 s
Leaderboard query (joins users + notes + likes)	~120 ms

Metric	Result
AI first-pass success on test set	~80%
Monthly operating cost	~\$8 USD

Visual prompt (Slide 10 — Performance Dashboard):

16:9 slide titled "Performance & Testing Results". Layout split into two regions horizontally with 60/40 ratio. Top banner across full width (in indigo #4F46E5 with white text, height ~80 px): "139 manual test cases, 14 test areas — all passed".

Left region (60% width) — KPI Circles: Three large concentric ring gauges side by side, each showing a Lighthouse-style score circle. From left to right:

- 1. "Landing Performance" — ring filled to 87/100 in green #16A34A, large "87" inside, label below
- 2. "Accessibility" — ring filled to 92/100 in green #16A34A, large "92" inside, label below
- 3. "AI First-Pass Success" — ring filled to 80/100 in rose #E11D48, large "80%" inside, label below

Right region (40% width) — Speed & Cost Stats: Vertical stack of 4 mini stat cards with white fill and slate left border:

- 1. Icon (lightning bolt) + "Initial HTML" / "~800 ms"
- 2. Icon (cursor) + "Time to Interactive" / "~1.5 s"
- 3. Icon (database) + "Leaderboard Query" / "~120 ms"
- 4. Icon (dollar) + "Monthly Operating Cost" / "~\$8 USD" (this card highlighted with light amber #FEF3C7 fill to make the cost stand out)

Bottom one-line takeaway banner: "Production-grade performance on a student budget." Flat vector, no gradients.

Slide 11 — Novelty of the Work ★

On slide:

What makes Prepify different — five user-facing differentiators:

#	Differentiator	What it means for the student
1	Built for Bangladesh	Faculty → Department → Batch → Course taxonomy + Bangla handwriting
2	Handwriting-first	Snap a notebook page — no typing, no scanning, no PDFs
3	Free, no credits	No paywall, no upload-to-download exchange, no subscription
4	AI explains your photo	Reads diagrams, equations, mixed scripts — not just text
5	A daily exam-prep habit	Streaks, badges, leaderboard, and curated collections

Other platforms cover one or two of these. Prepify is the first to combine all five.

Speaker notes: "Note sharing isn't new. Voting isn't new. AI explanations aren't new. What's new is the *combination* — and the fact that it's built around how Bangladeshi students actually study: photos of notebooks, shared in groups, organized by batch and course. Each of these five pieces individually exists somewhere; no platform offers all five together, and certainly not for free."

Visual prompt (Slide 11 — Five-Differentiator Strip):

16:9 slide titled "Novelty of the Work" in slate top-left. Five horizontal differentiator cards arranged as a single row across the middle 65% of the slide height, each card taking equal 19% of the slide width with 8 px gaps. Each card is a rounded rectangle (14 px corners) with white fill, 1.5 px slate border, and 20 px internal padding. Card layout top-to-bottom: (a) a 48-px filled colored circle at the top center containing a large monoline icon in white, (b) a bold short title (18 pt slate) on one or two lines, (c) a single one-line description in muted slate (12 pt) underneath.

Cards in order:

1. Indigo #4F46E5 circle, icon: Bangladesh map outline. Title: "Built for Bangladesh". Description: "Faculty → Department → Batch → Course"
2. Rose #E11D48 circle, icon: notebook with pencil. Title: "Handwriting-first". Description: "Snap a notebook page — no typing"
3. Emerald #059669 circle, icon: tag with \$0. Title: "Free, no credits". Description: "No paywall, no exchange model"
4. Amber #F59E0B circle, icon: sparkles over a document. Title: "AI explains your photo". Description: "Diagrams, equations, mixed scripts"
5. Blue #2563EB circle, icon: flame + medal. Title: "A daily habit". Description: "Streaks, badges, leaderboard, collections"

Below the strip of cards (lower 18% of slide), a single full-width banner with a light gray fill and a slate left border, containing one bold italic line centered in indigo #4F46E5 (22 pt): "*Other platforms cover one or two of these — Prepify is the first to combine all five.*"

Above the cards (top 12% of slide, just under the title), a thin one-line subtitle in muted slate (14 pt): "Uniqueness measured in user value, not technology checkboxes." Flat vector, no gradients, generous whitespace.

Slide 12 — Sample Dataset & Expected Output

On slide:

Input: Photographed handwritten solution (1–10 pages, JPEG/PNG/WebP, ≤10 MB each)

Example input: A 2-page handwritten solution to "Find the time complexity of merge sort using recurrence relation"

Output (generated by Gemini 2.5 Flash):

```
{
  "summary": "The notes derive the time complexity of merge sort
              by solving its recurrence  $T(n) = 2T(n/2) + n$ "
}
```

```

        using the master method, arriving at  $O(n \log n)$ .",
    "steps": [
      { "stepNumber": 1, "title": "Set up the recurrence",
        "content": "Express merge sort's cost as  $T(n) = 2T(n/2) + n...$ " },
      { "stepNumber": 2, "title": "Apply the master theorem",
        "content": "Identify  $a=2$ ,  $b=2$ ,  $f(n)=n$ . Compute  $\log_b(a)=1...$ " },
      { "stepNumber": 3, "title": "Conclude the bound",
        "content": "Since  $f(n) = \Theta(n^{\log_b(a)})$ , Case 2 applies  $\rightarrow T(n) = \Theta(n \log n)$ ."
    }
  ],
  "keyConcepts": [
    { "name": "Master Theorem",
      "description": "Solves divide-and-conquer recurrences..." },
    { "name": "Recurrence Relation",
      "description": "Mathematical expression for recursive cost..." }
  ],
  "observations": "Page 2 contains a small arithmetic slip in line 3 – the constant cancels.",
  "topics": ["Merge Sort", "Master Theorem", "Recurrence Relations",
    "Time Complexity", "Divide and Conquer"]
}

```

Visual prompt (Slide 12 — Input/Output Composite):

16:9 slide titled "Sample Input → Structured Output". Three-column layout with 35/10/55 width ratio.

Left column (35%) — header label "INPUT (handwritten)" in slate, then a realistic-looking mockup of a photographed handwritten notebook page with mathematical content showing the merge-sort recurrence $T(n) = 2T(n/2) + n$, master theorem cases, and ending with $T(n) = O(n \log n)$. The page should look like notebook paper with horizontal lines (cream/yellowish background #FFFBEB) and dark blue ink-style handwriting. Add a slight rotation (~2 degrees) for realism. Below the page, a small caption: "1 of 2 pages".

Center column (10%) — large vertical arrow in indigo #4F46E5 pointing right with the label "Gemini 2.5 Flash" running vertically along it.

Right column (55%) — header label "OUTPUT (structured JSON)" in slate, then a rendered AI-explanation panel mockup matching the actual Prepify UI:

- Top: "Summary" heading + 2 lines of summary text in light gray box
- Middle: 3 numbered step cards stacked, each with title in indigo and 1 line of body
- Below steps: 2 "Key Concept" definition rows (term in bold, description in regular)
- Below: small amber-bordered observation callout
- Bottom: row of 5 topic-tag chips in rose-bordered pill style

Add a small footer below the right column: "Validated by Zod schema, written to `ai_explanation` table". Flat vector, no gradients.

Slide 13 — Web Interface (Part 1)

On slide:

Two screenshots side by side:

- **Landing page** (light mode) — hero, stats strip, featured notes, department browser
- **Note detail page** (dark mode) — title, uploader, image gallery, comments

Caption: Responsive design — adapts from 375 px mobile to 1920 px desktop without horizontal scrolling.

Visual: Reuse Figure 3.6 (Landing + Note Detail) screenshot composite from your report, with a thin border around each capture and a presentation title bar above.

Slide 14 — Web Interface (Part 2 — AI + Dashboards)

On slide:

Three screenshots:

- **AI Explanation Panel** — summary + numbered steps + key concepts + topic chips
- **Moderator Dashboard** — pending queue with approve/reject controls
- **Admin Dashboard** — platform statistics + user role management

Visual: Reuse Figures 4.5 and 4.6 screenshot composites from your report. Tile them in a 2×2 grid (top: AI panel + moderator queue, bottom: admin dashboard + user management) with consistent borders and a small caption under each.

Slide 15 — Conclusion

On slide:**Delivered:**

- ☒ Complete platform — auth, notes, engagement, moderation, AI
- ☒ Multimodal AI pipeline with structured output and regen cap
- ☒ Localized taxonomy + gamification layer
- ☒ Production deployment at prepify.space
- ☒ All 3 project objectives met

Honest limitations:

- No automated test suite (manual testing only)
- Small active user base (not yet stress-tested)
- Image-only uploads (no PDF support yet)
- Email-only notifications (no Web Push)

Next steps:

- Automated quiz generation from notes (Phase III)
- Similar-note discovery via embedding the AI-generated topic tags
- PDF support, Web Push notifications, multi-institution rollout

Visual prompt (Slide 15 — Three-Column Conclusion):

16:9 slide titled "Conclusion". Three equal-width vertical columns separated by thin vertical dividers, each with a colored header strip at the top and a list of items below.

Left column — green #16A34A header strip with white text "DELIVERED". Below: 5 list items each prefixed with a green ☒ checkmark in 18-pt slate text. Items:

1. Complete platform — auth, notes, engagement, moderation, AI
2. Multimodal AI pipeline with structured output and regen cap
3. Localized taxonomy + gamification layer
4. Production deployment at [prepify.space](#)
5. All 3 project objectives met

Center column — amber #F59E0B header strip with white text "LIMITATIONS". Below: 4 list items each prefixed with a small amber  icon in 18-pt slate text. Items:

1. No automated test suite (manual testing only)
2. Small active user base (not yet stress-tested)
3. Image-only uploads (no PDF support yet)
4. Email-only notifications (no Web Push)

Right column — indigo #4F46E5 header strip with white text "NEXT STEPS". Below: 4 list items each prefixed with a small indigo → arrow in 18-pt slate text. Items:

1. Automated quiz generation from notes (Phase III)
2. Similar-note discovery via topic-tag embeddings
3. PDF support + Web Push notifications
4. Multi-institution rollout

Bottom of slide: a single one-line banner in muted gray italic: "*Honest about what's done, what's not, and what's next.*" Flat vector, no gradients.

Slide 16 — References

On slide (cite the most important 6–8 only — full IEEE list is in the report):

- [1] M. A. Chatti et al., "Toward a Personal Learning Environment Framework," *IJVPLE*, 2010.
- [2] G. Attwell, "Personal Learning Environments — the future of eLearning?," *eLearning Papers*, 2007.
- [3] S. Getenet et al., "Students' digital technology attitude and online learning engagement," *IJETHE*, 2024.
- [4] Google DeepMind, "Gemini: A Family of Highly Capable Multimodal Models," arXiv:2312.11805, 2023.
- [5] A. Vaswani et al., "Attention Is All You Need," *NeurIPS*, 2017.
- [6] R. Smith, "An overview of the Tesseract OCR engine," *ICDAR*, 2007.
- [7] Vercel, "Next.js Documentation," 2026.
- [8] Drizzle Team, "Drizzle ORM Documentation," 2026.

Visual: No new visual needed — plain reference list slide. Use a small "References" header in slate, then the 8 references in two columns (4 per column) with reference numbers in indigo.

Slide 17 — Thank You / Q&A

On slide:

- **Thank You**
- Questions?
- Live demo available at **`**prepify.space**`**
- Source code: GitHub (Shorno Kamal Roy)
- Contact: [your email]

Visual prompt (Slide 17 — Closing):

16:9 closing slide. White background. Centered: large "Thank You" in indigo #4F46E5 (96 pt). Below it: smaller "Questions?" in slate (40 pt). Further below: a clean info card (light gray fill, slate border, rounded corners) containing four lines centered:

- 🌐 Live demo: **prepify.space**
- 💻 Source: **GitHub — Shorno Kamal Roy**
- ✉️ Contact: **[your email]**
- 🎓 Daffodil International University, Department of CSE

Bottom of slide, in muted gray (16 pt): "Supervised by Md. Sazzadur Ahamed and Md. Ashraful Islam Talukder". No background images, no decorative borders, just clean professional layout.

APPENDIX — Backup Slides (Keep Ready for Q&A)

These don't appear in the main flow but should be in your deck just past Slide 17, ready to jump to if asked. Each one ~30 seconds to explain.

Backup A — ER Diagram

Visual: Reuse Figure 3.3 (ER Diagram) from `Diagram_Generation_Prompts.md`. Use for any "show me the schema" question.

Backup B — `generateExplanation` Server Action Code

On slide: Code snippet of the actual server action with the 5 validation checks highlighted. Use for any "how do you control AI cost / abuse" question.

Visual prompt (Backup B — Code Snippet with Highlights):

16:9 slide titled "Backup — `generateExplanation` Server Action". Centered: a code block on dark slate #0F172A background showing this TypeScript code with VSCode-like syntax highlighting:

```

export async function generateExplanation(noteId: number) {
  // 1. Auth check → 401
  const session = await checkAuth();
  if (!session?.user) return unauthorized();

  // 2. Note exists + load with course, files → 404
  const targetNote = await db.query.note.findFirst({
    where: eq(note.id, noteId),
    with: { course: true, files: true },
  });
  if (!targetNote) return notFound();

  // 3. Ownership check → 403
  if (targetNote.userId !== session.user.id) return forbidden();

  // 4. Status check → 400
  if (targetNote.status !== "approved") return badRequest();

  // 5. Regen cap → 429
  if (existing && existing.regenerateCount >= MAX_REGENERATIONS)
    return tooManyRequests();

  // → call Gemini 2.5 Flash with Zod schema
  const aiResult = await generateNoteExplanation(
    targetNote.files.map(f => f.url),
    targetNote.course.name
  );

  // → persist with status='ready'
  await db.update(aiExplanation)
    .set({ ...aiResult, status: "ready" })
    .where(eq(aiExplanation.id, recordId));
}

```

Highlight each numbered comment line (1-5) with a red left border bar in #DC2626. Add a small label on the right edge for each: "401", "404", "403", "400", "429" in matching red badges. Title at top: "Five validation gates protect cost and integrity." Flat code-editor aesthetic.

Backup C — Lighthouse Score Breakdown

On slide: Detailed Lighthouse breakdown — Performance / Accessibility / Best Practices / SEO with sub-metrics.

Visual prompt (Backup C — Lighthouse Detail):

16:9 slide titled "Backup — Lighthouse Score Breakdown". Recreate the actual Lighthouse score panel layout: four large ring gauges in a row (Performance, Accessibility, Best Practices, SEO) with scores (87, 92, 95, 90 respectively) in green if ≥ 90 or amber if 80-89. Below the gauges, a 2-column "Performance Metrics" table:

- First Contentful Paint: 0.8 s
- Largest Contentful Paint: 1.5 s
- Total Blocking Time: 60 ms
- Cumulative Layout Shift: 0.02
- Speed Index: 1.4 s
- Time to Interactive: 1.5 s

Below the table, a small note: "Measured on production deployment at prepify.space, mobile profile, fast 3G simulation". White background, flat vector.

Presentation Flow Tips

Aspect	Recommendation
Total time	12–15 minutes presentation + 3–5 minutes Q&A
Time per slide	~45 seconds average; spend ~2 minutes on Novelty (Slide 11) since it is the differentiator slide
Live demo	If allowed, do a 90-second live demo right after Slide 14 — log in, upload a note, generate an AI explanation in real time
Font sizing	Title 36–44 pt, bullets 24–28 pt, never below 18 pt
Visual ratio	At least 50% of every slide should be visual (figure / table / screenshot), not text
Backup slides	Three backup slides (A, B, C) ready to jump to if asked
Color rule	Stick to the palette in the Style Preface — don't introduce new colors
Animation rule	One simple fade transition between slides; avoid spinning, zooming, or bouncing animations

Tool Recommendations

Slide / Visual	Best tool to generate
Slide 1, 17	PowerPoint / Google Slides directly
Slides 2, 3, 6, 9, 11, 15 (cards / grids)	Figma or PowerPoint with shape tools
Slides 4, 5 (tables)	PowerPoint table styles
Slides 7, 8, 13, 14 (reused figures)	Already generated for the report

Slide / Visual	Best tool to generate
Slide 10 (gauges + KPI)	PowerPoint with native chart, or Canva templates
Slide 12 (input/output composite)	Take a real photo of handwritten notes for the left side, screenshot the AI panel for the right side, composite in Figma
Backup B (code)	Carbon (https://carbon.now.sh) with the "Slate" theme

For all AI image generators (ChatGPT, Gemini, Whimsical), prepend the **Common Visual Style Preface** at the top of this document to every prompt for consistency.